Math Inquirers

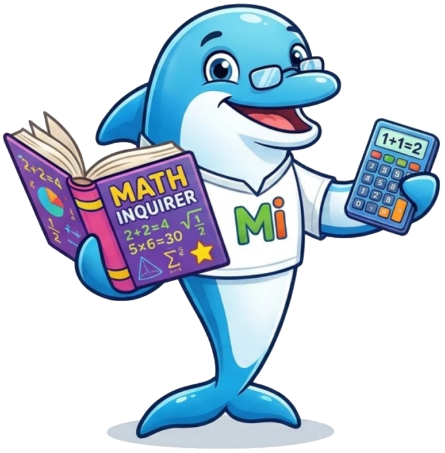
Think. Question. Prove.

Level 1 (ages 6-8) - Term 2

Workbook

Book 3

Lessons 11 to 15



My name:

Contents

Math Inquirers Level 1, Term 2 - Workbook, Book 3

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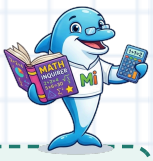
Lesson 15 **Equal Groups** page 35
Multiplication: Equal Groups

Kit pages (cut-outs)

Some lessons end with a kit page: tens bars, number lines, cards and so on.
Each kit page has a blank back, so you can cut or tear it out without losing any work.
Tear along the dashed line in the left margin, then cut out the pieces. Keep them in an envelope.

WHO IS WHERE?!

Name _____ Date _____



After question 3 you may choose: carry on, or try **Think harder**.

Use your things in a row. Say your answer out loud, then write it.

No.S1

Study the picture. The door is on the left.

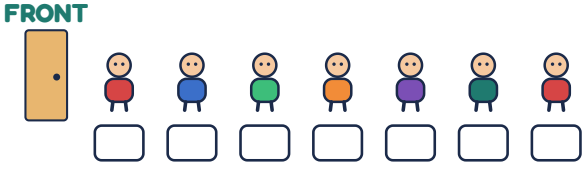


Statement	True	False
Wei is 3rd.	☐	☐
Jenny is 5th.	☐	☐
Lily is at the back.	☐	☐

Put a tick (✓) in the correct box.

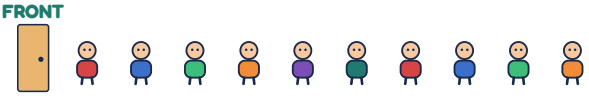
No.S2

Fill in the places of the children. The door is on the left.



No.S3

Count the children. The door is on the left.



(a) There are _____ children in the line.

(b) The last child is _____ in the line.

Name _____ Date _____

Look at the picture first. Then read the question.

No.S4

Aisha is 2nd in the line. Kumar stands right behind Aisha.



Kumar is _____ in the line.

No.S5

Jenny is 2nd from the front in a line of 6 children.



Jenny is _____ from the back.

No.S6

Wei is 1st. Lily is 3rd. Kumar stands between Wei and Lily.



Kumar is _____ in the line.

No.S7

Ben is 5th in the line.



There are ☐ children in front of Ben.

Name _____ Date _____

Draw the line or use your things in a row. Choose any question, in any order.

No.T1

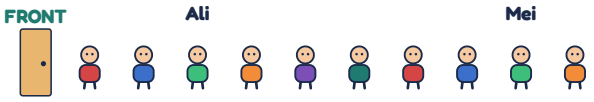
Mei is 5th from the front and 5th from the back.

(a) Kumar says there are 10 children in the line. Is Kumar correct? Circle Yes or No. **Yes** **No**

(b) There are _____ children in the line.

No.T2

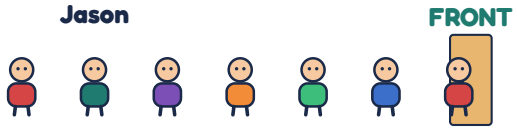
Ali is 3rd and Mei is 9th in a line of 10 children.



There are ☐ children between Ali and Mei.

No.T3

The door is on the right. Count the places from the door.



Jason is _____ in the line.

No.T4

There are 8 children in a line. Wei is 2nd from the front. Lily is 3rd from the back.

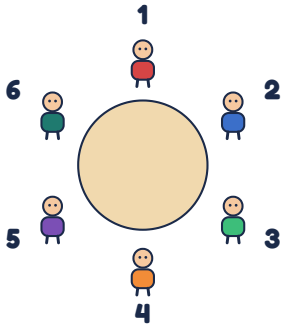
There are ☐ children between Wei and Lily.

Name _____ Date _____

Choose any question. You do not have to finish them all.

No.G1

Six children sit round a table. The seats are numbered 1 to 6. Ali is in seat 1, Mei is in seat 3 and Ben is in seat 5.



(a) Siti sits next to Ali and next to Mei. Siti is in seat _____.

(b) Ravi sits opposite Ali. Ravi is in seat _____.

(c) Kumar sits next to Ben and next to Ali. Kumar is in seat _____.

No.G2

A line has 13 children. Mei is 4th from the front. Ben is 3rd from the back.

(a) There are ☐ children between Mei and Ben.

(b) The middle child is _____ from the front.

No.G3

Four children ran a race. Siti was 1st. Ben beat Ali. Ali beat Mei.

(a) Ben was _____.

(b) Mei was _____.

Name _____ Date _____

Investigation: Two places for one child

Try this any time. You do not have to finish.

First, try this. Draw 5 dots in a line. Write 1st to 5th above the dots. Below the dots, write the places counting from the back.

Next, try this. Do the same for a line of 8 dots. Pick 3 children. For each, add the place from the front and the place from the back. I notice ...

Exit ticket

On your own, at the end of the lesson. No looking back.

No.E1

Ben is 4th in the line.

There are ☐ children in front of Ben.

No.E2

Mei is 5th from the front in a line of 6 children.

Mei is _____ from the back.

No.E3

Aisha is 3rd in the line. Kumar is right behind Aisha. Wei is right behind Kumar.

Wei is _____ in the line.

Look-back There are no right answers. You can pass. Nobody reads it out.

Something I did well:

The hardest part was ... ☐ saying 1st, 2nd, 3rd ☐ counting from the back ☐ 4 or 4th ☐ how many are between ☐ something else

What helped me: ☐ my things in a row ☐ touching each one ☐ drawing the line ☐ I asked for help ☐ I was not stuck

Next time I will try: ☐ use my things in a row ☐ draw the line ☐ check my answer ☐ ask a question ☐ I am still stuck (my grown-up will photograph this page for my tutor) ☐ I would like to pass

Name _____ Date _____

Kit page: place cards and number cards

A grown-up cuts these out before the lesson. The top two rows say which place. The bottom two rows say how many.

1st	2nd	3rd	4th	5th
6th	7th	8th	9th	10th
1	2	3	4	5
6	7	8	9	10

TEAR OUT here (blank on the back) - then cut out the pieces

BAR STRIPS: MORE AND FEWER!

Name _____ Date _____



After question 3 you may choose: carry on, or try **Think harder**.

Look at the strips. A longer strip means more. Say your answer out loud, then write it.

No.S1

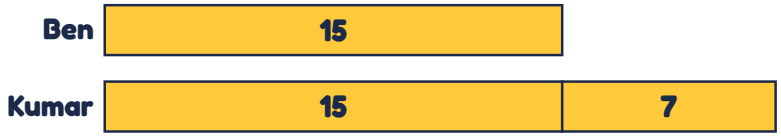
Ali has 12 stickers. Mei has 8 stickers.



Ali has ☐ more stickers than Mei.

No.S2

Ben has 15 marbles. Kumar has 7 more marbles than Ben.



Kumar has ☐☐ marbles.

No.S3

Siti has 34 beads. Dev has 17 more beads than Siti.



(a) Dev has ☐☐ beads.

(b) They have ☐☐ beads altogether.

Name _____ Date _____

Draw or use your strips if you like. Read each question carefully.

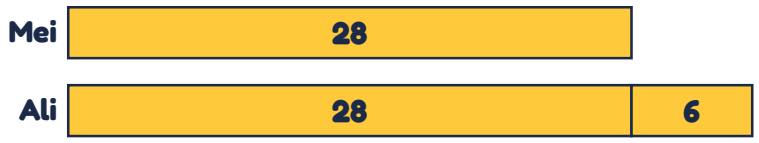
No.S4

Mei has 9 cards. Ravi has 5 cards. Draw a strip for Mei and a strip for Ravi.

Mei has ☐ more cards than Ravi.

No.S5

Mei has 28 stickers. Ali has 6 more stickers than Mei.

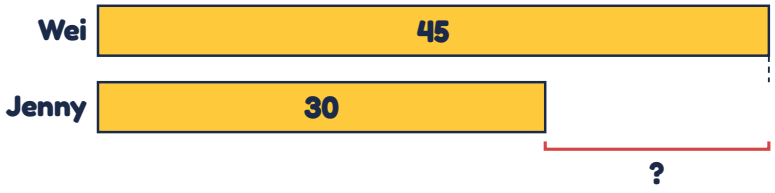


(a) Ali has ☐☐ stickers.

(b) They have ☐☐ stickers altogether.

No.S6

Wei has 45 stamps. Jenny has 30 stamps.



Wei has ☐☐ more stamps than Jenny.

No.S7

Aisha has 40 beads. Lily has 24 beads. Which number sentence finds how many more beads Aisha has than Lily? Write its number in the brackets.

(1) 24 + 40

(2) 40 - 24

(3) 40 + 40

(4) 24 + 24

( )

Name _____ Date _____

Use your strips or draw them. Choose any question, in any order.

No.T1

Ali and Mei have 40 stickers altogether. Ali has 8 more stickers than Mei.



(a) Mei has ☐☐ stickers.

(b) Ali has ☐☐ stickers.

No.T2

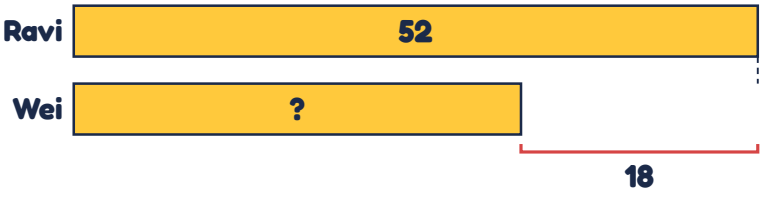
Ali has 30 stickers. Mei has 18 stickers. Ali gives some stickers to Mei. Then they have the same number.



Ali gives Mei ☐ stickers.

No.T3

Wei has 18 fewer marbles than Ravi. Ravi has 52 marbles.



(a) Wei has ☐☐ marbles.

(b) They have ☐☐ marbles altogether.

Name _____ Date _____

Choose any question. You do not have to finish them all.

No.G1

52 books are on two shelves. 7 books move from the top shelf to the bottom shelf. Then both shelves have the same number of books.

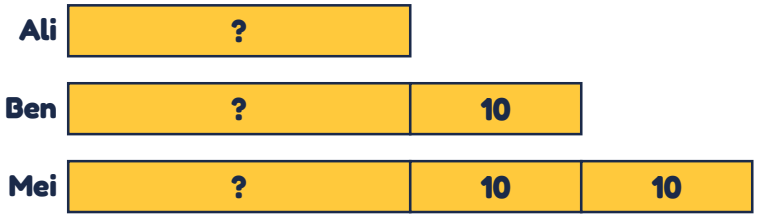


(a) At first the top shelf had ☐☐ books.

(b) At first the bottom shelf had ☐☐ books.

No.G2

Ali, Ben and Mei have 90 cards altogether. Ben has 10 more cards than Ali. Mei has 10 more cards than Ben.



(a) Ali has ☐☐ cards.

(b) Mei has ☐☐ cards.

No.G3

Kumar has 12 more marbles than Siti. Siti gives 4 marbles to Kumar.

Kumar has ☐☐ more marbles than Siti now.

Name _____ Date _____

Investigation: Same total, different gaps

Try this any time. You do not have to finish.

First, try this. Ali and Mei have 10 stickers altogether. Ali has 2 more than Mei. Draw two strips. How many stickers does each child have?

Next, try this. Now try the same total, 10, with 4 more, and then 6 more. Write each pair. Line 4: I notice ...

Exit ticket

On your own, at the end of the lesson. No looking back.

No.E1

Ali has 13 stickers. Mei has 9 stickers.

Ali has ☐ more stickers than Mei.

No.E2

Ben has 27 marbles. Siti has 15 more marbles than Ben.

Siti has ☐☐ marbles.

No.E3

Wei has 25 cards. Jenny has 8 more cards than Wei.

They have ☐☐ cards altogether.

Look-back There are no right answers. You can pass. Nobody reads it out.

Something I did well: .....

The hardest part was ... ☐ drawing the strips ☐ how many more ☐ more than or fewer than ☐ altogether ☐ something else

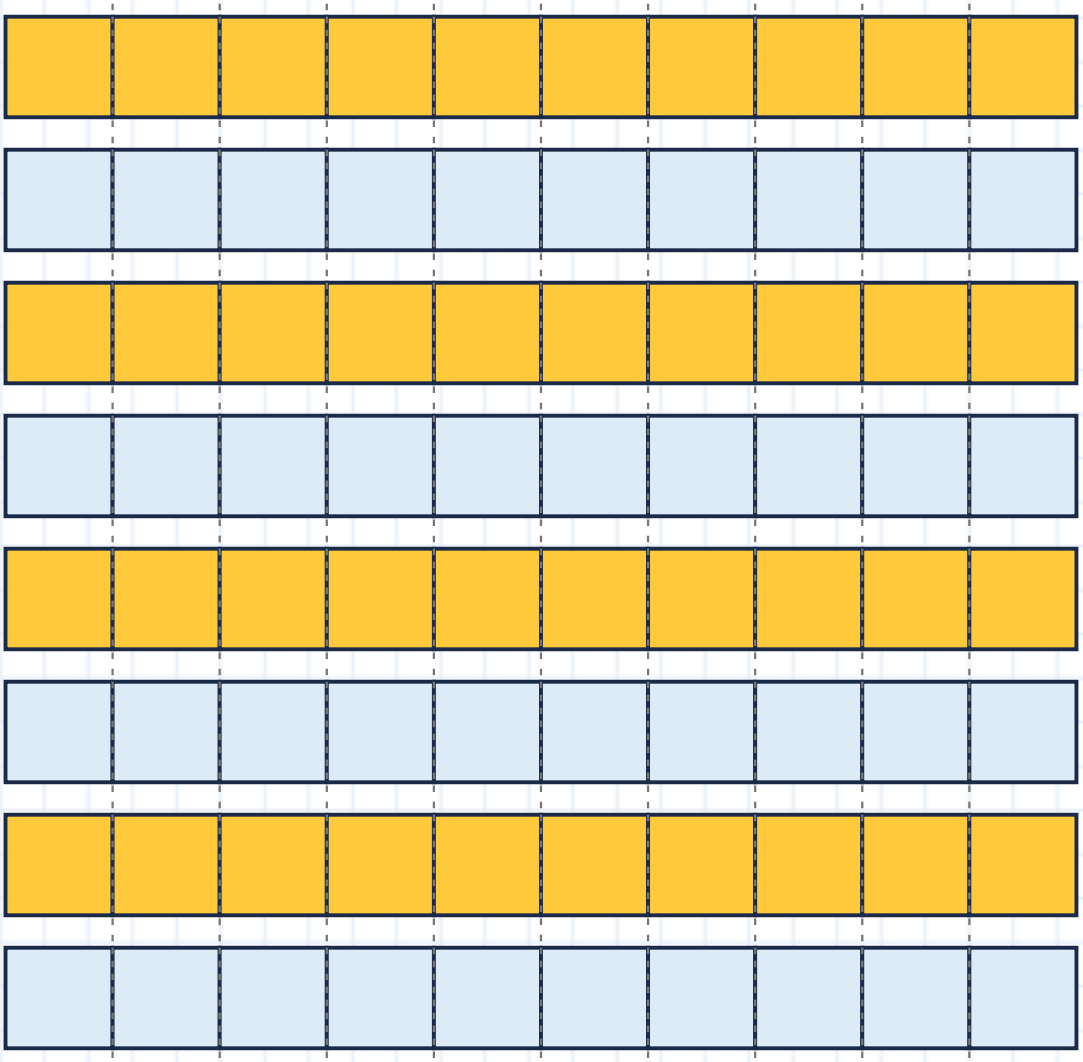
What helped me: ☐ my strips ☐ drawing boxes ☐ counting up ☐ I asked for help ☐ I was not stuck

Next time I will try: ☐ use my strips ☐ draw the strips ☐ check my answer ☐ ask a question ☐ I am still stuck (my grown-up will photograph this page for my tutor) ☐ I would like to pass

Name _____ Date _____

Kit page: strips

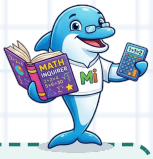
A grown-up cuts out the strips before the lesson. Cut a strip to any length at a dotted line.



TEAR OUT here (blank on the back) - then cut out the pieces

DOUBLES AND NEAR DOUBLES!

Name _____ Date _____

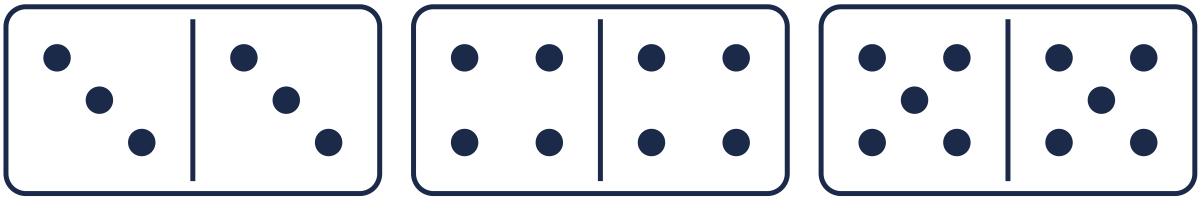


After question 3 you may choose: carry on, or try **Think harder**.

A double is two of the same number, like 2 + 2. Use counters or a drawing. Say your answer out loud, then write it.

No.S1

Look at the dominoes. Fill in the missing numbers.



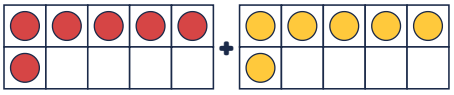
(a) 3 + 3 = ☐

(b) 4 + 4 = ☐

(c) 5 + 5 = ☐☐

No.S2

Mei has 6 red counters and 6 yellow counters. Fill in the missing numbers.



(a) 6 + 6 = ☐☐

(b) Double 6 is ☐☐.

No.S3

Fill in the missing numbers.

(a) Double 2 is ☐.

(b) Double 5 is ☐☐.

(c) Double 7 is ☐☐.

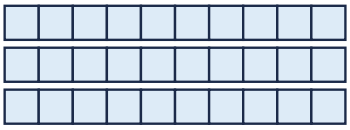
(d) Double 9 is ☐☐.

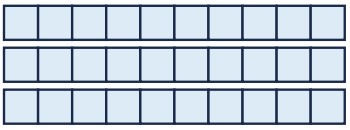
Name _____ Date _____

Use counters or the bars to help. Read each box carefully.

No.S4

Look at the bars. Fill in the missing numbers.





(a) 3 tens + 3 tens = ☐ tens

(b) 30 + 30 = ☐☐

(c) Double 40 is ☐☐.

No.S5

Ali has 8 stickers. Ben has 8 stickers. How many stickers do they have altogether?



(a) ☐ + ☐ = ☐☐

(b) They have _____ stickers altogether.

No.S6

Read each statement. Decide if it is True or False.

Statement	True	False
Double 4 is 8.	☐	☐
Double 6 is 14.	☐	☐
9 + 9 is a double.	☐	☐
Double 30 is 60.	☐	☐

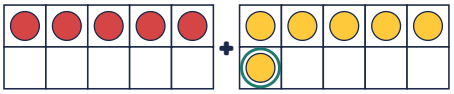
Put a tick (✓) in the correct box.

Name _____ Date _____

Use counters or the ten-frame page. Then fill in the answers. Choose any question, in any order.

No.T1

Look at the counters. The ringed counter is 1 more. Fill in the missing numbers.



(a) Double 5 is ☐☐.

(b) 5 + 6 = ☐☐

No.T2

Add.

(a) 3 + 4 = ☐

(b) 8 + 7 = ☐☐

(c) 10 + 11 = ☐☐

(d) 2 + 3 = ☐

No.T3

Which sum is not a near double? Write the number in the bracket.

(1) 6 + 7

(2) 8 + 9

(3) 5 + 8

(4) 9 + 10

( )

No.T4

Kumar adds 8 + 9. Step 1: 8 + 8 = 16. Step 2: 16 - 1 = 15. Answer (a) and (b).

(a) Is Kumar correct? Circle Yes or No. **Yes** **No**

(b) 8 + 9 = ☐☐

Name _____ Date _____

Choose any question. You do not have to finish them all.

No.G1

Add 8 + 9 + 8 + 7 + 8. Show your working on the lines.

The total is ☐☐.

No.G2

Ben says 9 + 10 + 11 is the same as 10 + 10 + 10. Answer (a) and (b). Show your working on the lines.

(a) Is Ben correct? Circle Yes or No. **Yes** **No**

(b) 19 + 20 + 21 = ☐☐

No.G3

Find double 35. Fill in the missing numbers. Show your working on the lines.

(a) Double 30 is ☐☐.

(b) Double 5 is ☐☐.

(c) Double 35 is ☐☐.

Name _____ Date _____

Investigation: Make two groups the same

Try this any time. You do not have to finish.

First, try this. Draw 6 dots and 8 dots. Move 1 dot from the 8 to the 6. What do you see?

Next, try this. Draw dots for each pair. Try 5 and 9, then 7 and 8. Can you move dots so both groups are the same? When does it work? Write what you notice.

Exit ticket

On your own, at the end of the lesson. No looking back.

No.E1

Fill in the missing number.

Double 9 is ☐☐.

No.E2

Fill in the missing number.

Double 30 is ☐☐.

No.E3

Add.

9 + 8 = ☐☐

Look-back There are no right answers. You can pass. Nobody reads it out.

Something I did well:

The hardest part was ... ☐ remembering the doubles ☐ adding 1 more or 1 less ☐ doubling tens like 30 + 30 ☐ choosing which double (Think harder) ☐ something else

What helped me: ☐ counters or ten-frames ☐ drawing it ☐ trying again ☐ I asked for help ☐ I was not stuck

Next time I will try: ☐ drawing it ☐ using objects ☐ checking my answer ☐ asking a question ☐ I am still stuck (my grown-up will photograph this page for my tutor)

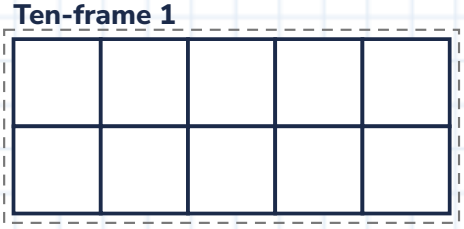
☐ I would like to pass

Name _____ Date _____

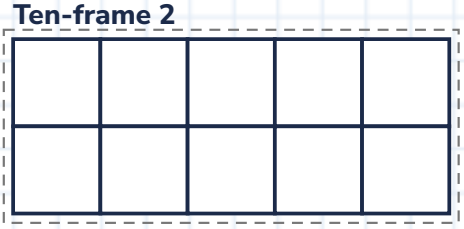
Kit page: ten-frames and counters

A grown-up cuts out the 20 counters (10 red, 10 yellow) before the lesson. Keep them in an envelope. The frames can stay on the page.

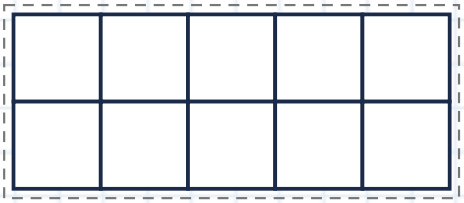
Ten-frame 1



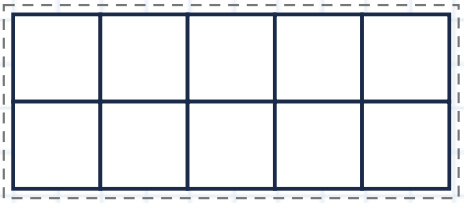
Ten-frame 2



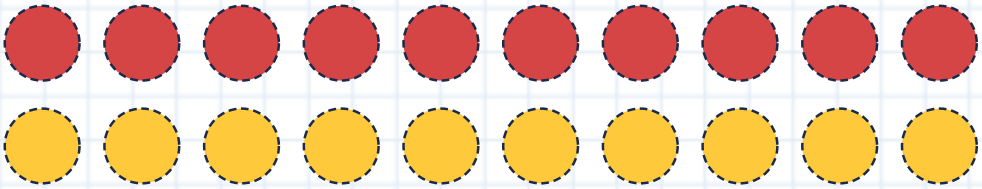
Ten-frame 3



Ten-frame 4



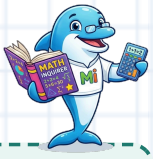
Counters: cut out all 20 (10 red, 10 yellow)



TEAR OUT here (blank on the back) - then cut out the pieces

ODD + ODD: ALWAYS, SOMETIMES, NEVER!

Name _____ Date _____



After question 3 you may choose: carry on, or try **Think harder**.

A number is even if you can make pairs with none left over. It is odd if one is left over. Use counters or draw dots in pairs.

No.S1

Look at the pairs. Fill in the missing numbers.



(a) 6 counters: ☐ left over.

(b) 7 counters: ☐ left over.

No.S2

Pair up counters or draw dots in pairs. Circle Odd or Even for each number.

(a) 4 **Odd** **Even**

(b) 9 **Odd** **Even**

(c) 10 **Odd** **Even**

(d) 13 **Odd** **Even**

No.S3

Count in 2s. Colour each number you say. Then fill in the missing number.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20

The last digits are 2, 4, 6, 8 and ☐.

Name _____ Date _____

Use your counters. Read each box carefully.

No.S4

Which number is even? Write its number in the bracket.

(1) 7

(2) 11

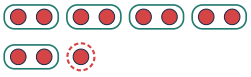
(3) 14

(4) 15

( )

No.S5

Look at the 11 counters in pairs. Fill in the blank. Then circle Odd or Even.



(a) ☐ counter has no partner.

(b) 11 is **Odd** **Even**

No.S6

Read each statement. Decide if it is True or False.

Statement	True	False
6 is even.	☐	☐
9 is even.	☐	☐
15 is odd.	☐	☐
20 is odd.	☐	☐

Put a tick (✓) in the correct box.

Name _____ Date _____

Use counters or the number cards. Test each claim. Choose any question.

No.T1

Mei says: an odd number plus an even number is always odd. Test it. Fill in the blanks.

(a) 6 + 5 = ☐☐. Circle Odd or Even. **Odd** **Even**

(b) 10 + 3 = ☐☐. Circle Odd or Even. **Odd** **Even**

(c) 8 + 1 = ☐. Circle Odd or Even. **Odd** **Even**

(d) Does the claim work for all three sums? Circle Yes or No. **Yes** **No**

No.T2

Circle Odd or Even for each number.

(a) 57 **Odd** **Even**

(b) 80 **Odd** **Even**

(c) 34 **Odd** **Even**

(d) 91 **Odd** **Even**

No.T3

Ravi says: an odd number plus an odd number is always odd. Test it with 9 + 3. Fill in the blanks.

(a) 9 + 3 = ☐☐

(b) Is Ravi correct? Circle Yes or No. **Yes** **No**

(c) An odd number plus an odd number is odd. Circle Always, Sometimes or Never. **Always** **Sometimes** **Never**

No.T4

Can you make each number by adding two odd numbers? Circle Yes or No.

(a) 12 **Yes** **No**

(b) 15 **Yes** **No**

(c) 20 **Yes** **No**

Name _____ Date _____

Choose any question. You do not have to finish them all.

No.G1

Look at 3 + 5. Draw 3 counters and 5 counters in pairs. Fill in the missing numbers.

(a) 3 counters have ☐ left over.

(b) 5 counters have ☐ left over.

(c) The two left-over counters make ☐ pair.

No.G2

Add three odd numbers. Circle Odd or Even for each total.

(a) 3 + 5 + 7 = ☐☐. Circle Odd or Even. **Odd** **Even**

(b) 1 + 1 + 1 = ☐. Circle Odd or Even. **Odd** **Even**

(c) 9 + 3 + 5 = ☐☐. Circle Odd or Even. **Odd** **Even**

(d) What kind of number is odd + odd + odd? Circle Odd or Even. **Odd** **Even**

No.G3

Look at 4 + 6. Draw 4 counters and 6 counters in pairs. Fill in the blank, then finish the sentence.

(a) ☐ counters are left over.

(b) Even + even is even because _____.

Name _____ Date _____

Investigation: Numbers next to each other

Try this any time. You do not have to finish.

First, try this. Add 3 + 4. Is the total odd or even?

Next, try this. Add 5 + 6, 8 + 9 and 10 + 11 too. Write odd or even for each total. Is the total always odd, sometimes odd or never odd? Write what you notice.

Exit ticket

On your own, at the end of the lesson. No looking back.

No.E1

Is 17 odd or even? Circle Odd or Even.

17 is **Odd** **Even**

No.E2

Which number is even? Write its number in the bracket.

(1) 9

(2) 13

(3) 18

(4) 21

( )

No.E3

Can you make 14 by adding two odd numbers? Circle Yes or No.

14 **Yes** **No**

Look-back There are no right answers. You can pass. Nobody reads it out.

Something I did well: _____

The hardest part was ... ☐ telling odd from even ☐ testing the sums ☐ always, sometimes or never ☐ drawing the pairs (Go further) ☐ something else

What helped me: ☐ counters in pairs ☐ drawing it ☐ trying again ☐ I asked for help ☐ I was not stuck

Next time I will try: ☐ drawing it ☐ using objects ☐ checking my answer ☐ asking a question ☐ I am still stuck (my grown-up will photograph this page for my tutor)

☐ I would like to pass

Name _____ Date _____

Kit page: number cards 1 to 20

A grown-up cuts out the 20 cards before the lesson. Keep them in an envelope.

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20

TEAR OUT here (blank on the back) - then cut out the pieces

EQUAL GROUPS!

Name _____ Date _____

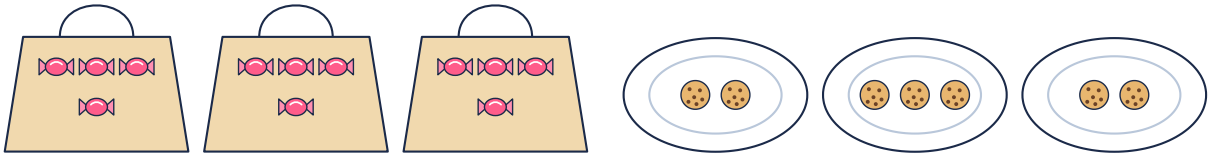


After question 3 you may choose: carry on, or try **Think harder**.

Use counters and plates, or draw it.

No.S1

Are the groups equal? Circle Yes or No for each picture.

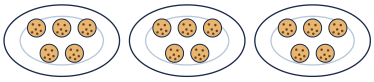


(a) Left picture: **Yes** **No**

(b) Right picture: **Yes** **No**

No.S2

Look at the picture. Fill in the blanks.



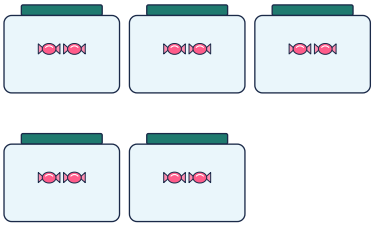
(a) There are ☐ plates.

(b) There are ☐ cookies on each plate.

(c) ☐ groups of ☐ = ☐☐

No.S3

Mei has 5 jars. Each jar has 2 sweets. How many sweets does Mei have in all?



(a) 2 + 2 + 2 + 2 + 2 = ☐☐

(b) 5 groups of 2 = ☐☐

Name _____ Date _____

Draw it, count it, then write it.

No.S4

Draw 4 dots on each plate. Then fill in the blank.



(a) 3 groups of 4 = ☐☐

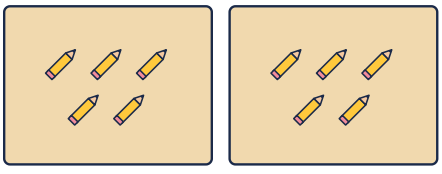
No.S5

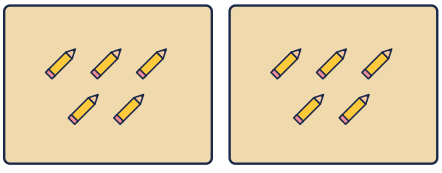
Fill in the table.

Groups	In each	Total
3	5	☐☐
4	2	☐
2	5	☐☐

No.S6

Aisha has 4 boxes. Each box has 5 pencils. How many pencils does Aisha have in all?





(a) Write a number sentence. ☐ ◯ ☐ = ☐

(b) Aisha has ☐☐ pencils in all.

Name _____ Date _____

Use counters or draw dots in rows. Choose any question, in any order.

No.T1

Look at the dots in rows. Fill in the blanks.



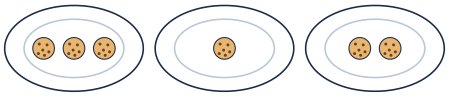
(a) Rows of dots: ☐ × ☐ = ☐☐

(b) Turn the picture on its side. ☐ × ☐ = ☐☐

(c) Is 2 × 8 = 8 × 2? Circle Yes or No. **Yes** **No**

No.T2

Each plate should have 3 cookies. Look at the picture. How many more cookies are needed to make the groups equal?



☐ more cookies are needed.

No.T3

Kumar has 3 plates with 4 apples on each plate. He says he has 7 apples.

(a) 3 groups of 4 = ☐☐

(b) Is Kumar correct? Circle Yes or No. **Yes** **No**

No.T4

Jenny has 12 stickers. She puts them into equal groups in different ways. Fill in the table.

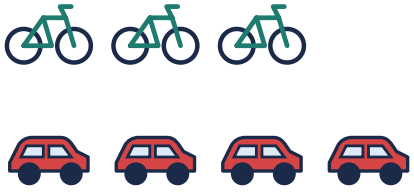
Groups	In each	Total
6	2	12
4	☐	12
☐	4	12

Name _____ Date _____

Choose any question. You do not have to finish them all.

No.G1

A shed has 3 bikes and 4 cars. Each bike has 2 wheels. Each car has 4 wheels. How many wheels are there in all?



(a) 3 bikes have ☐ wheels.

(b) 4 cars have ☐☐ wheels.

(c) There are ☐☐ wheels in all.

No.G2

Jamie has 6 jars. Each jar has 5 marbles. Jamie needs 40 marbles. How many more marbles does he need?

(a) 6 groups of 5 = ☐☐

(b) Jamie needs ☐☐ more marbles.

No.G3

Wei has 24 sweets. Show 3 different ways to put them into equal groups. Use a different number of groups each time. Many answers are right.

(a) ☐☐ groups of ☐☐ = 24

(b) ☐☐ groups of ☐☐ = 24

(c) ☐☐ groups of ☐☐ = 24

Name _____ Date _____

Investigation: Rows with 12 counters

First, try this. Put 12 counters in 2 rows of 6. Turn the grid on its side. How many rows now? Write the number on the first line.

Next, try this. Make rows and columns with 12 counters in other ways, such as 3 rows of 4. Write a × sentence for each on the lines. Try 5 rows. Does it work?

Exit ticket

On your own, at the end of the lesson. No looking back.

No.E1

Fill in the blank.

(a) 5 + 5 + 5 + 5 = ☐☐

No.E2

Which is the same as 4 groups of 3? Write its number in the bracket.

(1) 4 + 3

(2) 3 + 3 + 3 + 3

(3) 3 + 3 + 3

(4) 4 + 4 + 4 + 4

( )

No.E3

Ali has 3 boxes. Each box has 4 pencils. How many pencils does Ali have in all?

(a) Write a number sentence. ☐ ◯ ☐ = ☐

(b) Ali has ☐☐ pencils in all.

Look-back There are no right answers. You can pass. Nobody reads it out.

Something I did well: _____

The hardest part was ... ☐ telling equal from not equal ☐ counting the groups ☐ writing the × sentence ☐ something else

What helped me: ☐ counters on plates ☐ drawing it ☐ counting in jumps ☐ I was not stuck

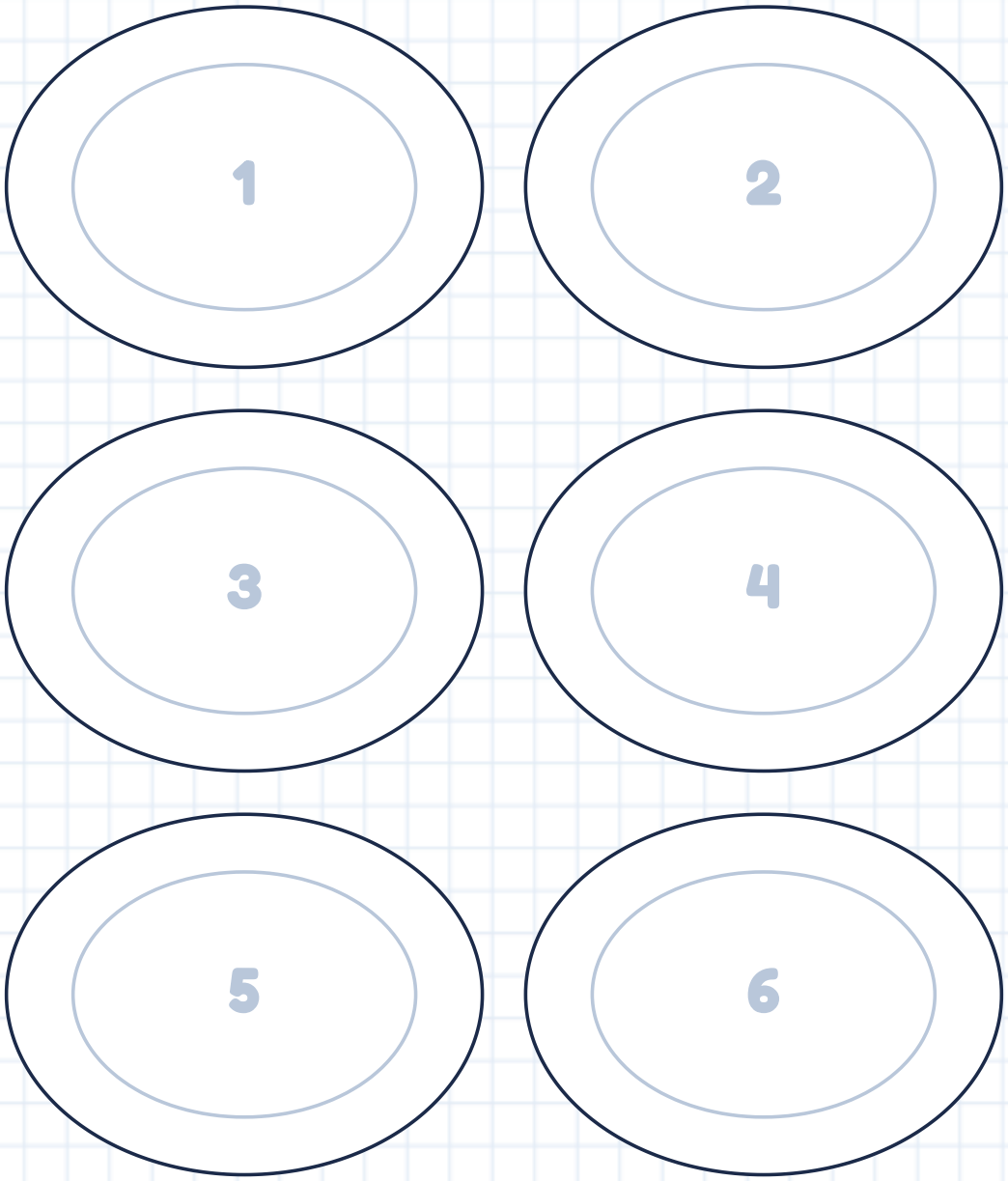
Next time I will try: ☐ drawing it ☐ using objects ☐ checking my answer ☐ asking a question ☐ I am still stuck (my grown-up will photograph this page for my tutor)

☐ I would like to pass

Name _____ Date _____

Kit page: groups mat

Nothing to cut. Lay counters on the plates. Use it if you do not have plates or bowls.

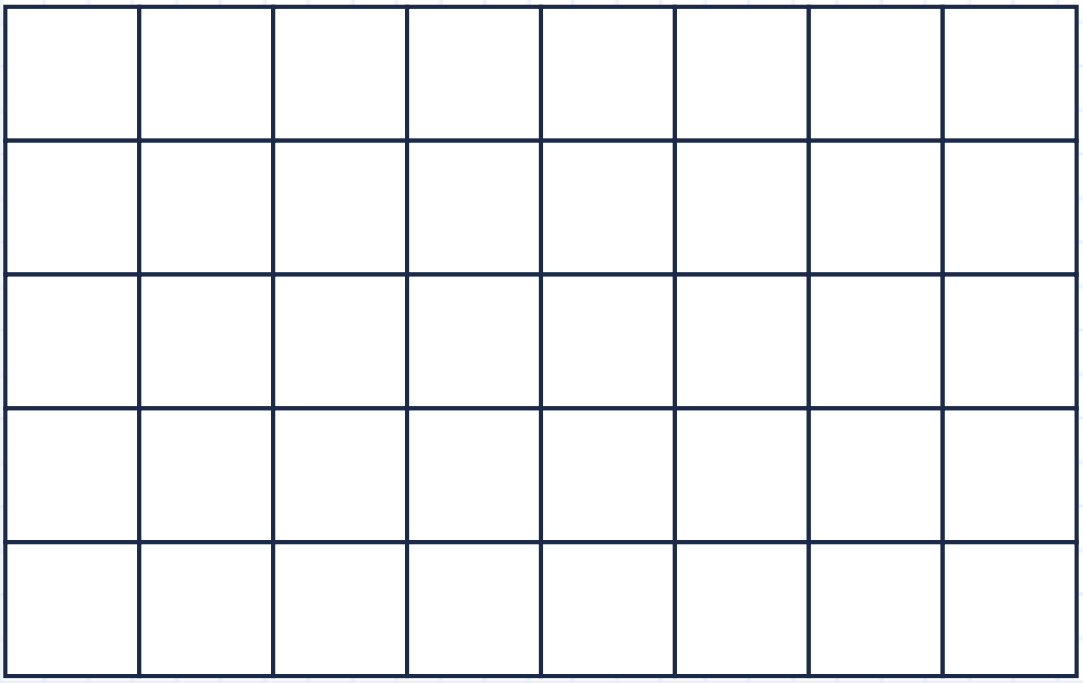


TEAR OUT here (blank on the back) - then cut out the pieces

Name _____ Date _____

Kit page: array mat

Nothing to cut. Lay counters in rows and columns on the squares.



TEAR OUT here (blank on the back) - then cut out the pieces